Overview

This project studies how a bank can improve the efficiency of a term deposit marketing campaign using machine learning. I use the Bank Marketing dataset from the UCI Machine Learning Repository and focus on the prediction track. The project combines a subscription prediction model with a separate call-duration model to support business targeting decisions under time and cost constraints.

Business Motivation

The decision-maker in this project is a bank marketing manager who must decide which clients to contact in a term deposit campaign. Contacting every client is inefficient because each call consumes staff time and generates labor cost, while many contacted clients do not subscribe. A good targeting strategy can improve campaign efficiency by concentrating outreach on more promising clients. In this project, the key business tradeoff is between the expected revenue from successful subscriptions and the time and cost required to reach clients.

Dataset Description

I use the Bank Marketing dataset from the UCI Machine Learning Repository. The dataset contains 41,188 observations and 21 columns, where each observation represents one client contact record from a Portuguese bank's direct marketing campaign. The target variable is y, which indicates whether the client subscribed to a term deposit. Important features include demographic variables such as age, job, marital status, and education, as well as campaign-related variables such as contact type, month, campaign, previous contacts, and prior campaign outcome. I also use the duration variable in a separate regression model to approximate expected call cost, but I exclude observed duration from the subscription classifier because it is only known after the call takes place.

Modeling Approach

To model the bank's targeting problem realistically, I separate prediction from optimization. For subscription prediction, I use only features that are available before the call takes place, such as age, job, marital status, education, contact type, month, campaign history, and prior campaign outcome. I exclude the observed duration variable from the subscription classifier because call duration is only known after the call and would create data leakage in a pre-call decision setting. Instead, I build a separate model for expected call duration and use that prediction in the optimization stage as a cost proxy.

I first train a logistic regression model as a baseline classifier because it is interpretable and provides a simple benchmark. I then compare it with a random forest classifier to capture nonlinear relationships and feature interactions. Because the downstream business decision depends on expected value calculations, probability calibration is also important. I therefore calibrate the random forest model and compare it with the baseline using ROC-AUC, average precision, F1-score, and Brier score. The calibrated random forest is selected as the final subscription model because it provides the best overall combination of ranking quality and probability quality.

To incorporate time cost into the business decision, I build a separate duration model using only pre-call features. I compare a linear regression baseline with a random forest regressor using a log-transformed duration target. Duration proves much harder to predict than subscription probability, so I treat the duration model as an approximate expected-cost

estimate rather than a precise forecast. Even so, it is still useful for testing whether a duration-aware targeting strategy can improve campaign efficiency under tighter operational constraints.

Results

The subscription prediction results show that the calibrated random forest performs best overall. On the validation set, it achieves a ROC-AUC of about 0.801, an average precision of about 0.466, and the best Brier score among the tested classifiers, indicating that it is the most suitable model for downstream expected-profit calculations. By comparison, the uncalibrated random forest improves recall but produces less reliable probabilities, while logistic regression gives more stable probabilities than the uncalibrated forest but slightly weaker ranking performance.

The duration models are much weaker than the subscription model, which is consistent with the idea that call length depends on many factors that are not fully observable before the call. On the validation set, the random forest regressor performs slightly better than the linear regression baseline, so I use it in the optimization stage. However, I interpret duration prediction cautiously and treat it as a rough estimate of expected call cost rather than an exact forecast.

I then compare several targeting policies, including contacting nobody, contacting everyone, random targeting under a time budget, probability-only targeting, and duration-aware targeting. On the held-out test set under the baseline business scenario (R = 100, c0 = 1, c1_per_min = 4, and a 3000-minute budget), both model-based policies strongly outperform naive baselines such as contacting everyone or selecting clients randomly. In this baseline setting, Top Probability Under Budget performs best, with a realized profit of about 26,950, compared with about 26,204 for Duration-Aware Under Budget. This suggests that under normal operating conditions, predicted subscription probability is the most important driver of campaign profitability.

At the same time, sensitivity analysis shows that this conclusion is not universal. When the call-time budget becomes tighter or the operational importance of call duration increases, the duration-aware policy becomes more competitive and can outperform probability-only targeting. On the held-out test set in a tighter-capacity stress scenario (R = 100, c0 = 1, c1_per_min = 1, and a 1500-minute budget), Duration-Aware Under Budget achieves a realized profit of about 22,081, exceeding the 21,319 achieved by Top Probability Under Budget. This result supports the instructor feedback that call duration can matter as a cost dimension and shows that the best targeting rule depends on business conditions rather than being fixed in all scenarios.

A final implementation concern is budget overrun risk. Because call duration must be predicted rather than known in advance, a policy that satisfies the predicted time budget may still exceed the realized time budget. In the baseline test scenario, both model-based policies exceed the actual 3000-minute budget, which suggests that a practical deployment should consider a safety buffer when selecting clients.

Business Recommendation and Impact

The main business recommendation is that the bank should not contact every client and should not rely on random outreach. Instead, it should use a model-based targeting strategy.

Under the baseline scenario, the strongest policy is to prioritize clients by predicted subscription probability, because this strategy delivers the highest realized profit on the held-out test set. In the baseline test case, Top Probability Under Budget achieves a realized profit of about 26,950, compared with a large loss from contacting everyone and a loss from random targeting. This shows that machine-learning-based targeting can substantially improve the efficiency of the campaign.

At the same time, the results also suggest that a single targeting rule may not be optimal under all business conditions. When call-center time becomes tighter, the value of incorporating expected call duration increases. In the tighter-capacity stress scenario, the Duration-Aware Under Budget policy achieves a realized profit of about 22,081, compared with about 21,319 for the probability-only policy. Therefore, my recommendation is conditional: in normal operating conditions, the bank should prioritize clients by predicted subscription probability, but under tighter staffing or time constraints, it should adopt a duration-aware targeting rule that accounts for both expected conversion value and expected call duration.

Interpretation for a Non-Technical Audience

This project shows that the bank can make better marketing decisions by being selective about whom it contacts. Rather than calling everyone, the bank should focus on clients who are more likely to subscribe to a term deposit. In normal conditions, the most profitable approach is to rank clients by their likelihood of subscribing. However, when staff time is limited, it can also be helpful to consider how long each call is expected to last, because long calls use up more resources. In simple terms, the model helps the bank spend its calling time more efficiently.

Limitations and Potential Bias

One important limitation is that expected call duration is difficult to predict using only pre-call information. Many factors that affect call length, such as the client's mood, availability, or the quality of the live interaction, are not observable before the call. As a result, the duration model should be treated as an approximate cost estimate rather than a precise forecast.

Another limitation is that the dataset is based on historical marketing campaigns from a specific banking context. Customer behavior, marketing channels, and macroeconomic conditions may differ from current conditions, so the exact profit numbers may not generalize perfectly to today's banking environment. In addition, the business impact calculations depend on assumed values for revenue per subscription and call cost. These assumptions are useful for comparing strategies, but they are not the same as real internal bank accounting data.

A further operational risk is that predicted-budget policies may still exceed the realized time budget because duration must be estimated rather than known in advance. This means that a bank implementing such a system should consider using a safety buffer to reduce the risk of overspending call-center capacity.

A potential source of bias is that historical campaign data may reflect past targeting decisions. If the bank previously contacted some types of clients more often than others, the

model may inherit those historical patterns. One possible mitigation step would be to validate the final targeting policy with a prospective pilot or A/B test before full deployment.

References

UCI Machine Learning Repository. Bank Marketing Dataset.
https://archive.ics.uci.edu/dataset/222/bank+marketing
Scikit-learn documentation for logistic regression, random forest, calibration, and evaluation metrics.

AI Use Declaration

I used ChatGPT to brainstorm the project structure, refine the business framing, assist with code debugging and organization, and improve the clarity of the written report. All final modeling decisions, analysis, interpretations, and submitted materials were reviewed and completed by me.